

Human nutrition

Physical digestion: The breakdown of food into smaller pieces without chemical change to the food molecule.

Types of teeth:

1- Incisors: Position in mouth: Front

Function: Biting off pieces of food

2- Canine: Position in mouth: Either side of incisors

Function: Tearing the food

3- Premolar: Position: Behind canines

Function: Chewing and grinding food

4- Molar: Position: Back of the mouth

Function: Chewing and grinding food

Structure of teeth:

Enamel: A hard substance covering the crown of the teeth

Dentine: Under the enamel and it contains layers of living cytoplasm.

Pulp cavity: Contains nerves and blood vessels that supply the dentine with nutrients and oxygen.

Cement: Covers the root of the tooth and it has fibers that attach the root to the jawbone

Tooth decay

- 1- Bacteria use the sugars left on teeth in anaerobic respiration
- 2- Anaerobic respiration produces an acid
- 3- Bacteria, sugars and acids accumulate on teeth to form the plaque which will harden overtime to form the tartar.
- 4- The acid produced by bacteria will dissolve the enamel of teeth, causing ~~bad~~^{tooth} decay.
- 5- Plaque might cause gum diseases as bacteria may infect gums, causing the tooth to become loose.

How to prevent tooth decay?

- * Eat less sugars
- * Use Fluoride tooth paste or Fluoride drinking water.
- * Brush teeth regularly
- * Visit the dentist regularly
- * Use antibacterial mouthwash and dental floss

Fluoride drinking water

Advantages:

- * Prevents tooth decay by strengthening the enamel
- * Treats the whole population
- * Free

Disadvantages:

- * Causes yellow stains on teeth
- * Might be toxic
- * Causes allergies
- * Bad taste in water
- * No individual choice

The mouth:

- Food is crushed by teeth and mixed with saliva
- The tongue mixes the food with saliva to form the bolus which is easier to swallow
- Saliva is produced from the salivary glands and it's a mixture of water, mucus, and amylase
- * Amylase catalyses the breakdown of starch into maltose.
- * Mucus lubricates the food and allows a bolus to form
- * Water allows the soluble parts of the food to dissolve

The esophagus:

- Delivers food from the mouth to the stomach
- Food is prevented from going down the trachea by the epiglottis, which blocks the trachea during swallowing
- Food moves along the esophagus by peristalsis

Peristalsis: The contraction and relaxation of muscles to push the food along.

- Circular muscles contract, and longitudinal muscles relax as they are antagonistic muscles. This makes the lumen of the alimentary canal narrower, pushing the food forward

The stomach:

- The stomach contains strong muscular walls that carry out peristalsis to mix the food with pepsin and HCl, to create an acidic mixture called the chyme.
- Proteins are broken down in the stomach in a reaction catalysed by the enzyme pepsin.
- HCl is produced to kill microbes in food and to provide pepsin an optimum pH.
- The thick stomach walls and the goblet cells in the stomach produce mucus to protect the stomach walls from being eaten away by HCl.

* Renin is an enzyme found in the stomach of young mammals. It catalyses the process of the clumping of milk so that it can be later broken down by the protease enzyme.

The small intestine: Duodenum:

- Walls of the small intestine produce enzymes to aid in digestion.
- Pancreatic juice, produced by the pancreas is delivered to the duodenum by the pancreatic duct.

The pancreatic juice contains:

- 1- Trypsin: Catalyses the breakdown of proteins into amino acids.
 - 2- Maltase: Catalyses the breakdown of maltose into glucose
 - 3- Lipase: Catalyses the breakdown of lipids into 3 fatty acids and glycerol
 - 4- Sodium hydrogen carbonate: An alkali that neutralizes the acidity of the chyme
- The bile, a yellow liquid produced in the liver and stored in the gallbladder, travels to the duodenum through the bile duct.
 - Bile helps in the digestion of fats. Bile salts break down fats into smaller droplets in emulsification. Which is a type of mechanical breakdown.

Emulsification: Breaking down of fats into small droplets with large surface area. which makes it easier for enzymes to work on fats.

The small intestine

Ileum

- The walls of the small intestine are covered with millions of tiny projections called villi.

- These villi are covered with millions of finger-like projections called microvilli.

* Absorption takes place through villi.

- Villi contain blood capillaries which allow amino acids, vitamins, minerals, salts and simple sugars to move into the blood by diffusion. Glucose moves into the blood by active transport.

- Villi contain the lacteal, which allows fatty acids to be transported to cells by the lymph fluid.

- After nutrients are absorbed into the blood, blood goes to the liver by the hepatic portal vein.

Adaptations of the villi:

- 1- Covered in microvilli to provide a larger surface area for absorption.
- 2- Contain a good supply of blood capillaries to allow efficient absorption.
- 3- Walls of the villi are only one cell thick to reduce the distance for diffusion of digested food into the blood.

Large intestine

- * Reabsorbs water from undigested food
- * Rectum stores the faeces
- * Anus allows undigested food to pass out of the body by egestion.

Diarrhoea

- It is the loss of water through faeces
- * Treatment could include oral rehydration therapy
- Diarrhoea is due to the bacterial infection called cholera.
- The bacteria lives and reproduce in the small intestine
- It produces a toxin that stimulates the cells in the lining of the intestine to secrete chloride ions which accumulate in the lumen of the small intestine.
- Water moves by osmosis from the blood into the lumen

Ingestion: The taking in of substances by the mouth into the body

Digestion: The break down of food

Absorption: The movement of nutrients from the intestine into the blood.

Assimilation: The uptake and use of nutrients.

Egestion: The removal of undigested food from the body as faeces.

Balanced diet: A diet that contains the correct proportions of the 7 nutrients to maintain good health.

1- Carbohydrates: Obtained from sweets, bread, rice, pasta.
Use in body: Storage, a source of energy.

2- Fats and oils: Obtained from oil and butter
Use in body: a source of energy, insulation against heat loss, for hormones, for cell membranes, insulation of nerve fibers.

3- Proteins: Obtained from fish, eggs and meat
Uses: Growth, tissue repair, enzymes, some hormones, cell membranes, hair, nails, can be broken down for energy.

4- Vitamins: Organic substances needed in very small amounts
* Vitamin C: Obtained from citrus foods; needed for healthy gums
Deficiency: Scurvy, which leads to bleeding gums

* Vitamin D: Made by the skin when exposed to sunlight, needed to help calcium absorption in bones
Deficiency: Rickets, which causes deformed bones

5- Minerals: Inorganic substances needed in small amounts
* Calcium: Obtained from dairy products; needed for strong bones and teeth.
Deficiency: Weak bones and teeth.

* Iron: Obtained from red meat and liver; needed to make hemoglobin and strong blood clotting.
Deficiency: Anemia; not enough oxygen is carried to cells, and poor blood clotting

6- Fiber: Obtained from fruits, vegetables, wheat, barley. Needed to prevent colon cancer and constipation, and helps peristalsis.
Deficiency: Colon cancer, constipation.

7- Water: Needed as a solvent, medium for chemical reactions, and to transport substances around the body



- Obesity is caused when people eat alot more fats than needed
- Obesity causes diabetes, strokes, heart attacks, and joints problems
- Malnutrition is caused when people have an unbalanced diets.

Food additives

Advantages:

- Longer shelf life
- Reduce growth of microbes
- Prevent food poisoning
- Improves the taste of food
- Improves the appearance of food by giving it color
- Emulsify (stabilize) food

Disadvantages:

- Hyperactivity in children
- Allergies
- Headaches and vomiting
- Asthma
- Possible links with cancer